RPM PACAKAGE MANAGEMENT

STEP:1

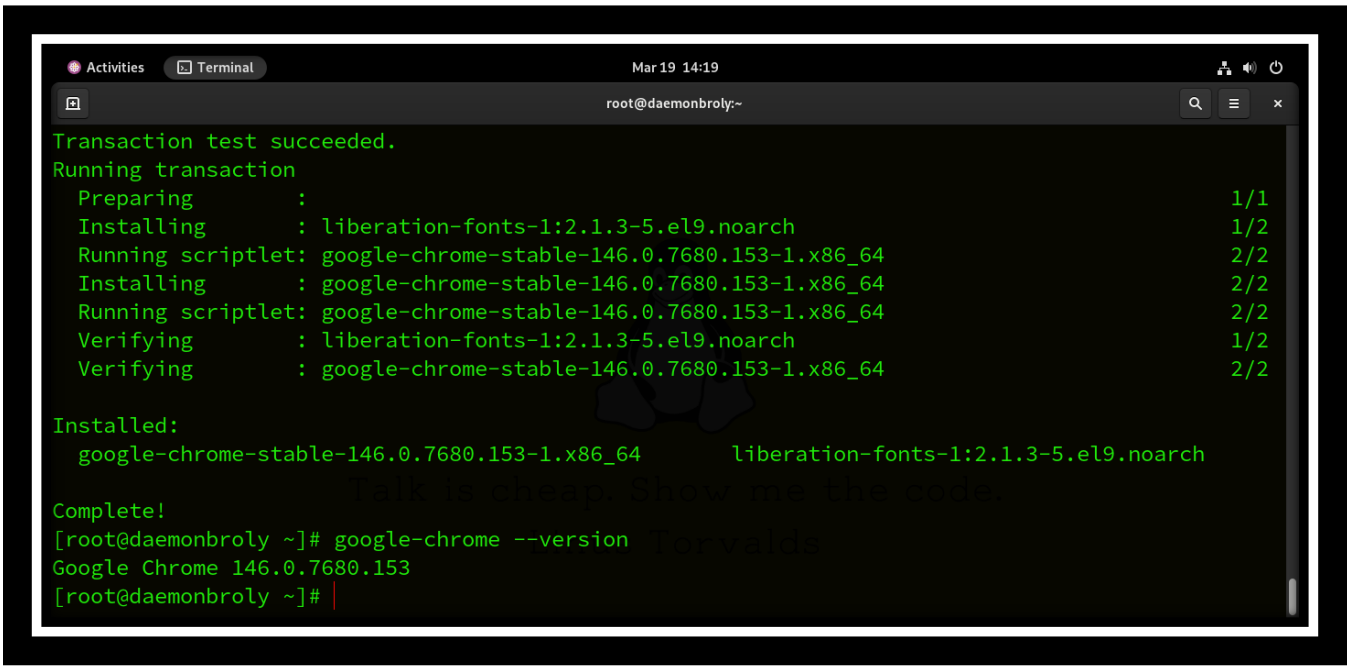
ADD URL

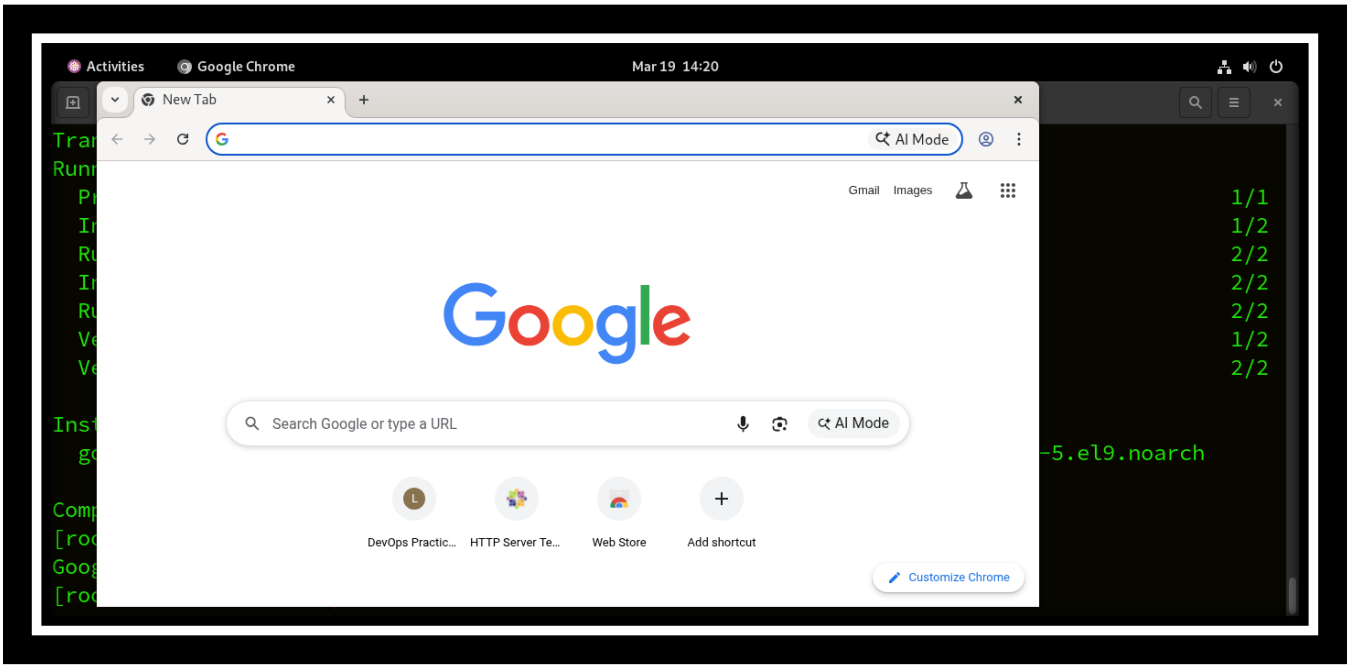
```
Activities   Terminal                    Mar 19  14:01
root@daemonbroly:/etc/yum.repos.d
[root@daemonbroly ~]# dnf config-manager --add-repo="https://dl.google.com/linux/direct/google-c
hrome-stable_current_x86_64.rpm"
Adding repo from: https://dl.google.com/linux/direct/google-chrome-stable_current_x86_64.rpm
[root@daemonbroly ~]# cd /etc/yum.repos.d/
[root@daemonbroly yum.repos.d]# ls
centos-addons.repo                                                  epel-next-testing.repo
centos.repo                                                         epel.repo
dl.google.com_linux_direct_google-chrome-stable_current_x86_64.rpm.repo  epel-testing.repo
epel-cisco-openh264.repo                                            google-chrome.repo
epel-next.repo
[root@daemonbroly yum.repos.d]#
```

VERFIY AND INSTALL

```
Activities   Terminal                    Mar 19  14:17
root@daemonbroly:~
Adding repo from: https://dl.google.com/linux/chrome/rpm/stable/x86_64
[root@daemonbroly ~]# cat /etc/yum.repos.d/dl.google.com_linux_chrome_rpm_stable_x86_64.repo
[dl.google.com_linux_chrome_rpm_stable_x86_64]
name=created by dnf config-manager from https://dl.google.com/linux/chrome/rpm/stable/x86_64
baseurl=https://dl.google.com/linux/chrome/rpm/stable/x86_64
enabled=1
[root@daemonbroly ~]# dnf install google-chrome-stable -y
CentOS Stream 9 - BaseOS                                  1.5 MB/s | 8.9 MB     00:06
CentOS Stream 9 - AppStream                               630 kB/s |  27 MB     00:44
CentOS Stream 9 - Extras packages                          13 kB/s |  20 kB     00:01
created by dnf config-manager from https://dl.google.com/linux/ 7.4 kB/s | 6.5 kB     00:00
Extra Packages for Enterprise Linux 9 - x86_64            963 kB/s |  20 MB     00:21
Extra Packages for Enterprise Linux 9 openh264 (From Cisco) - x 1.1 kB/s | 2.5 kB     00:02
Extra Packages for Enterprise Linux 9 - Next - x86_64      67 kB/s | 259 kB     00:03
Dependencies resolved.
================================================================================
 Package            Arch   Version         Repository                          Size
```

INSTALL SUCCESSFULLY





DISK MANAGEMENT

CREATING FDISK PARTION:

```
[root@daemonbroly ~]# lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sr0          11:0    1 1024M  0 rom
nvme0n1     259:0    0   20G  0 disk
├─nvme0n1p1 259:1    0    1G  0 part /boot
└─nvme0n1p2 259:2    0   19G  0 part
  ├─cs-root 253:0    0   17G  0 lvm  /
  └─cs-swap 253:1    0    2G  0 lvm  [SWAP]
nvme0n2     259:3    0    8G  0 disk
[root@daemonbroly ~]# fdisk /dev/nvme0n2

Welcome to fdisk (util-linux 2.37.4).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.


Command (m for help):
```

EXECUTING PRIMARY PARTITION:

```
Command (m for help): n
Partition type
   p   primary (0 primary, 0 extended, 4 free)
   e   extended (container for logical partitions)
Select (default p): p
Partition number (1-4, default 1): 2
First sector (2048-16777215, default 2048):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (2048-16777215, default 16777215): +3G

Created a new partition 2 of type 'Linux' and of size 3 GiB.

Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.

[root@daemonbroly ~]#
```

CRETAING FILE SYSTEM:

```
Activities  Terminal                Mar 19 14:46
root@daemonbroly:~
  ├─cs-root 253:0    0   17G  0 lvm  /
  └─cs-swap 253:1    0    2G  0 lvm  [SWAP]
nvme0n2     259:3    0    8G  0 disk
└─nvme0n2p2 259:4    0    3G  0 part
[root@daemonbroly ~]# mkfs.ext4 /dev/nvme0n2p2
mke2fs 1.46.5 (30-Dec-2021)
Found a dos partition table in /dev/nvme0n2p2
Proceed anyway? (y,N) y
Creating filesystem with 786432 4k blocks and 196608 inodes
Filesystem UUID: be9ced04-0750-4183-ad18-ca33018c7e4f
Superblock backups stored on blocks:
        32768, 98304, 163840, 229376, 294912

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done
```

MOUNTING THE DISK:

```
Activities  Terminal                Mar 19 14:49
root@daemonbroly:~
[root@daemonbroly ~]# lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sr0          11:0    1 1024M  0 rom
nvme0n1     259:0    0   20G  0 disk
├─nvme0n1p1 259:1    0    1G  0 part /boot
└─nvme0n1p2 259:2    0   19G  0 part
  ├─cs-root 253:0    0   17G  0 lvm  /
  └─cs-swap 253:1    0    2G  0 lvm  [SWAP]
nvme0n2     259:3    0    8G  0 disk
└─nvme0n2p2 259:4    0    3G  0 part /root/hello1
[root@daemonbroly ~]# mount /dev/nvme0n2p2 hello1
mount: /root/hello1: /dev/nvme0n2p2 already mounted on /root/hello1.
[root@daemonbroly ~]# df -T | tail -n 1
/dev/nvme0n2p2      ext4       3021608      24   2847916   1% /root/hello1
[root@daemonbroly ~]# blkid /dev/nvme0n2p2
/dev/nvme0n2p2: UUID="be9ced04-0750-4183-ad18-ca33018c7e4f" TYPE="ext4" PARTUUID="1c0307bb-02"
[root@daemonbroly ~]#
```

FSTAB ENTYRY:

```
#
# /etc/fstab
# Created by anaconda on Mon Nov 17 14:59:20 2025
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
#
# After editing this file, run 'systemctl daemon-reload' to update systemd
# units generated from this file.
#
/dev/mapper/cs-root     /                       xfs     defaults        0 0
UUID=b94d8b9e-9c55-46e8-88f7-8758ac44108c /boot                   xfs     defaults        0 0
/dev/mapper/cs-swap     none                    swap    defaults        0       0
UUID=be9ced04-0750-4183-ad18-ca33018c7e4f /hello1 ext4  defaults        0       0
~
-- INSERT --                                                            15,68-82      All
```

OUTPUT:

```
[root@daemonbroly ~]# blkid /dev/nvme0n2p2
/dev/nvme0n2p2: UUID="be9ced04-0750-4183-ad18-ca33018c7e4f" TYPE="ext4" PARTUUID="1c0307bb-02"
[root@daemonbroly ~]# vi /etc/fstab
[root@daemonbroly ~]# lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sr0          11:0    1 1024M  0 rom
nvme0n1     259:0    0   20G  0 disk
├─nvme0n1p1 259:1    0    1G  0 part /boot
└─nvme0n1p2 259:2    0   19G  0 part
  ├─cs-root 253:0    0   17G  0 lvm  /
  └─cs-swap 253:1    0    2G  0 lvm  [SWAP]
nvme0n2     259:3    0    8G  0 disk
└─nvme0n2p2 259:4    0    3G  0 part /root/hello1
[root@daemonbroly ~]#
```

LVM MANAGEMENT

PVCREATE :

```
Activities   Terminal                    Mar 19 15:01
root@daemonbroly:~
  └─cs-swap 253:1    0    2G  0 lvm  [SWAP]
nvme0n2     259:3    0    4G  0 disk
nvme0n3     259:4    0    4G  0 disk
[root@daemonbroly ~]# pvcreate /dev/nvme0n2
  Physical volume "/dev/nvme0n2" successfully created.
[root@daemonbroly ~]# pvs
  PV             VG Fmt  Attr PSize   PFree
  /dev/nvme0n1p2 cs lvm2 a--  <19.00g    0
  /dev/nvme0n2      lvm2 ---    4.00g 4.00g
[root@daemonbroly ~]# pvcreate /dev/nvme0n3
  Physical volume "/dev/nvme0n3" successfully created.
[root@daemonbroly ~]# pvs
  PV             VG Fmt  Attr PSize   PFree
  /dev/nvme0n1p2 cs lvm2 a--  <19.00g    0
  /dev/nvme0n2      lvm2 ---    4.00g 4.00g
  /dev/nvme0n3      lvm2 ---    4.00g 4.00g
[root@daemonbroly ~]#
```

VGCREATE, LV CREATE:

```
Activities   Terminal                    Mar 19 15:05
root@daemonbroly:~
  Physical Volume "lvpart1" not found in Volume Group "part1".
[root@daemonbroly ~]# lvs
  LV   VG Attr       LSize   Pool Origin Data%  Meta%  Move Log Cpy%Sync Convert
  root cs -wi-ao---- <17.00g
  swap cs -wi-ao----   2.00g
[root@daemonbroly ~]# lvcreate -L +2G -n part2 part1
  Logical volume "part2" created.
[root@daemonbroly ~]# lvs
  LV    VG    Attr       LSize   Pool Origin Data%  Meta%  Move Log Cpy%Sync Convert
  root  cs    -wi-ao---- <17.00g
  swap  cs    -wi-ao----   2.00g
  part2 part1 -wi-a-----   2.00g
[root@daemonbroly ~]# vgs
  VG    #PV #LV #SN Attr   VSize   VFree
  cs      1   2   0 wz--n- <19.00g    0
  part1   1   1   0 wz--n-  <4.00g <2.00g
[root@daemonbroly ~]#
```

LVEXTEND:

```
Activities   Terminal                    Mar 19  15:08
                          root@daemonbroly:~
  VG    #PV #LV #SN Attr   VSize   VFree
  cs      1   2   0 wz--n- <19.00g     0
  part1   1   1   0 wz--n-  <4.00g <2.00g
[root@daemonbroly ~]# lvextent -L +1G /dev/part1/part2
bash: lvextent: command not found...
[root@daemonbroly ~]# lvextend -L +1G /dev/part1/part2
  Size of logical volume part1/part2 changed from 2.00 GiB (512 extents) to 3.00 GiB (768 extent
s).
  Logical volume part1/part2 successfully resized.
[root@daemonbroly ~]# lv
bash: lv: command not found...
[root@daemonbroly ~]# lvs
  LV    VG    Attr       LSize   Pool Origin Data%  Meta%  Move Log Cpy%Sync Convert
  root  cs    -wi-ao---- <17.00g
  swap  cs    -wi-ao----   2.00g
  part2 part1 -wi-a-----   3.00g
[root@daemonbroly ~]#
```

PARITITION CRETAED :

```
Activities   Terminal                    Mar 19  15:11
                          root@daemonbroly:~
         =                       sectsz=512   sunit=0 blks, lazy-count=1
realtime =none                   extsz=4096   blocks=0, rtextents=0
[root@daemonbroly ~]# mount /dev/part1/part2 hello1
[root@daemonbroly ~]# lsblk
NAME          MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sr0            11:0    1 1024M  0 rom
nvme0n1       259:0    0   20G  0 disk
├─nvme0n1p1   259:1    0    1G  0 part /boot
└─nvme0n1p2   259:2    0   19G  0 part
  ├─cs-root   253:0    0   17G  0 lvm  /
  └─cs-swap   253:1    0    2G  0 lvm  [SWAP]
nvme0n2       259:3    0    4G  0 disk
└─part1-part2 253:2    0    3G  0 lvm  /root/hello1
nvme0n3       259:4    0    4G  0 disk
[root@daemonbroly ~]# blkid /dev/part1/part2
/dev/part1/part2: UUID="e5cd1a2d-e0f7-4049-9a47-6add2f723016" TYPE="xfs"
[root@daemonbroly ~]#
```

CREATING SWAP USING LVM

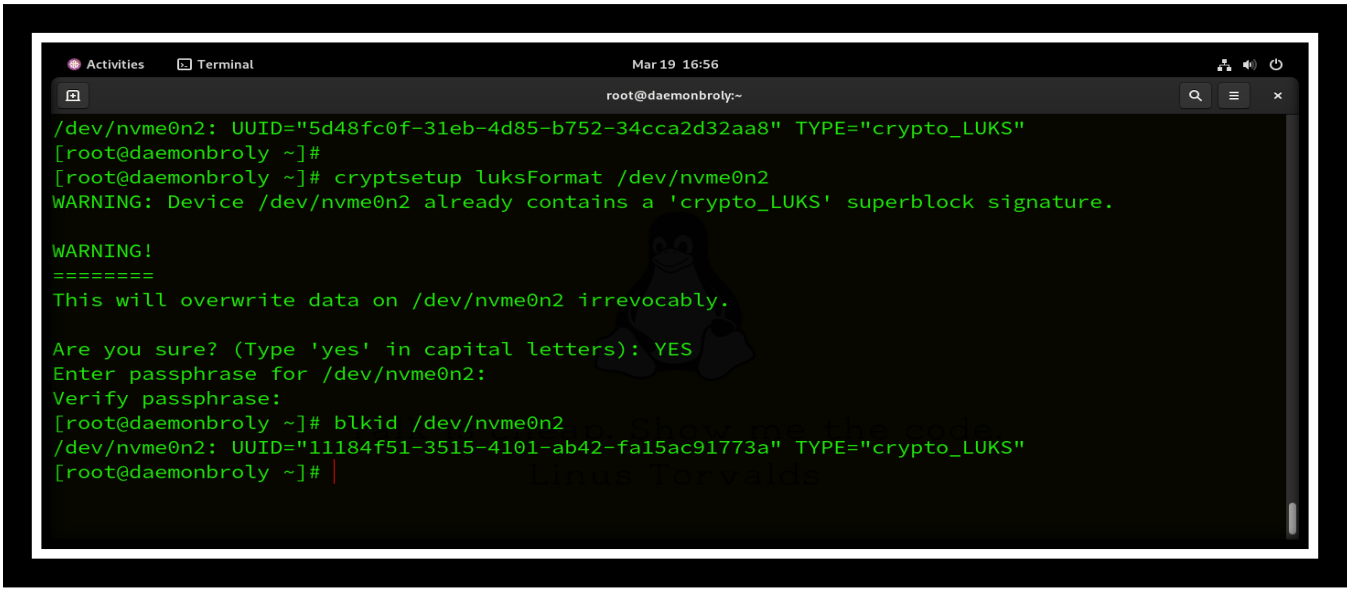
```
Activities  Terminal                Mar 19 16:42
root@daemonbroly:~
[root@daemonbroly ~]# lvs
  LV   VG    Attr       LSize   Pool Origin Data%  Meta%  Move Log Cpy%Sync Convert
  root cs    -wi-ao---- <17.00g
  swap cs    -wi-ao----   2.00g
  swap part1 -wi-a-----   2.00g
[root@daemonbroly ~]# mkswap /dev/part1/swap
Setting up swapspace version 1, size = 2 GiB (2147479552 bytes)
no label, UUID=56525596-a4ad-4574-961c-b3d6d61aeff0
[root@daemonbroly ~]# swapon /dev/part1/swap
[root@daemonbroly ~]# vi /etc/fstab
[root@daemonbroly ~]# lsblk
NAME         MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sr0           11:0    1 1024M  0 rom
nvme0n1      259:0    0   20G  0 disk
├─nvme0n1p1  259:1    0    1G  0 part /boot
└─nvme0n1p2  259:2    0   19G  0 part
  ├─cs-root  253:0    0   17G  0 lvm  /
```

```
Activities  Terminal                Mar 19 16:41
root@daemonbroly:~
Setting up swapspace version 1, size = 2 GiB (2147479552 bytes)
no label, UUID=56525596-a4ad-4574-961c-b3d6d61aeff0
[root@daemonbroly ~]# swapon /dev/part1/swap
[root@daemonbroly ~]# vi /etc/fstab
[root@daemonbroly ~]# lsblk
NAME         MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sr0           11:0    1 1024M  0 rom
nvme0n1      259:0    0   20G  0 disk
├─nvme0n1p1  259:1    0    1G  0 part /boot
└─nvme0n1p2  259:2    0   19G  0 part
  ├─cs-root  253:0    0   17G  0 lvm  /
  └─cs-swap  253:1    0    2G  0 lvm  [SWAP]
nvme0n2      259:3    0    4G  0 disk
└─part1-swap 253:2    0    2G  0 lvm  [SWAP]
nvme0n3      259:4    0    4G  0 disk
[root@daemonbroly ~]# vi /etc/fstab
[root@daemonbroly ~]#
```

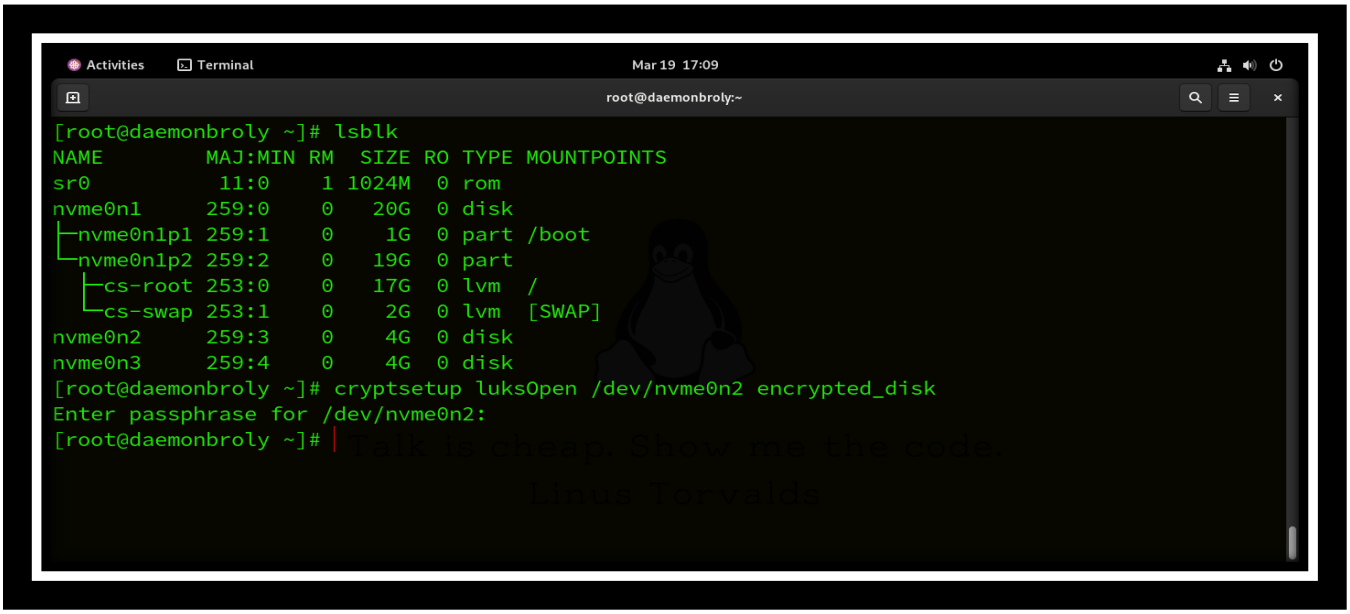
PERFORMING DM-CRYPT FOR BLOCK LEVEL ENCRYPTION

WARNING!! BEFORE FORMAT BACKUP THE DATA

STEP:1



STEP 2:



STEP 3:

```
realtime =none                   extsz=4096   blocks=0, rtextents=0
[root@daemonbroly ~]# ls
1.jpg  anaconda-ks.cfg  file2     hello   helloworld  myprofile   shared
1.sh   file1            file.jpg  hello1  msql.db     service.sh  world
[root@daemonbroly ~]# mount /dev/mapper/encrypted_disk hello1
[root@daemonbroly ~]# lsblk
NAME             MAJ:MIN RM  SIZE RO TYPE  MOUNTPOINTS
sr0               11:0    1 1024M  0 rom
nvme0n1          259:0    0   20G  0 disk
├─nvme0n1p1      259:1    0    1G  0 part  /boot
└─nvme0n1p2      259:2    0   19G  0 part
  ├─cs-root      253:0    0   17G  0 lvm   /
  └─cs-swap      253:1    0    2G  0 lvm   [SWAP]
nvme0n2          259:3    0    4G  0 disk
└─encrypted_disk 253:2    0    4G  0 crypt /root/hello1
nvme0n3          259:4    0    4G  0 disk
[root@daemonbroly ~]#
```

STEP 4:

```
[root@daemonbroly ~]# umount hello1
[root@daemonbroly ~]# cryptsetup luksClose encrypted_disk
[root@daemonbroly ~]# lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sr0          11:0    1 1024M  0 rom
nvme0n1     259:0    0   20G  0 disk
├─nvme0n1p1 259:1    0    1G  0 part /boot
└─nvme0n1p2 259:2    0   19G  0 part
  ├─cs-root 253:0    0   17G  0 lvm  /
  └─cs-swap 253:1    0    2G  0 lvm  [SWAP]
nvme0n2     259:3    0    4G  0 disk
nvme0n3     259:4    0    4G  0 disk
[root@daemonbroly ~]#
```